BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 194 · Floating-point representation and normalisation

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Write −2 in 8-bit two's complement.

Q2. What is unsigned 00100000 after a logical right shift by one?

Q3. Convert binary 0.1_2 to decimal.

Q4. Calculate 0.5×2^3.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

mantissa — The signed significant part of a floating-point number.

exponent — The power of two scaling the mantissa.

normalised — A two's-complement fractional mantissa with different first two bits.

precision — Number of significant bits represented.

range — Spread of magnitudes that can be represented.

New learning

Why is 0.0110 not normalised? — For a positive two's-complement fractional mantissa, a normalised value begins with 01, not 00.

Shift and compensate — Shift 0.0110 left one place to 0.1100. Decrease the exponent from 4 to 3 to compensate.

Verify the value — 0.0110_2 is 0.375; 0.375 × 16 = 6. 0.1100_2 is 0.75; 0.75 × 8 = 6.

Worked example

Normalise positive $0.0011_2 \times 2^5$.

1. First shift — 0.0011 becomes 0.0110; reduce exponent 5 → 4. It still begins 00.

2. Second shift — 0.0110 becomes 0.1100; reduce exponent 4 → 3. The first two bits are now 01.

3. Check — Both represent 6 in decimal, so normalisation preserved the value.

We Do · Practise with guidance

Normalise positive $0.0101_2 \times 2^4$.

```
0.0101 × 2⁴
→ ? × 2?
```

Predict the shift

Shift the mantissa left one position: 0.0101 → 0.1010.

Adjust and check

Exponent decreases from 4 to 3. Both representations equal decimal 5.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Identify mantissa and exponent in $0.1100_2 \times 2^3$.

Question 2

Normalise positive $0.0010_2 \times 2^5$.

Question 3

Why can normalisation improve precision without increasing the number of mantissa bits?

Exit Ticket · Answer before checking

Shifting 0.0110×2^4 left to 0.1100×2^4 preserves its value.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. 11111110.

Q2. 00010000.

Q3. 0.5.

Q4. 4.

You Do pages · Independent questions

Q1. Mantissa 0.1100_2; exponent 3.

Q2. $0.0010 \times 2^5 \rightarrow 0.0100 \times 2^4 \rightarrow 0.1000 \times 2^3$.

Q3. Redundant leading sign bits are removed so available mantissa positions hold more significant information. Exponent adjustment preserves the represented value.

Exit Ticket · Model answer

False. To preserve the value, the exponent must decrease to 3.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: